

Section A

Answer **all** the questions in this Section in the spaces provided.

1	(a)		Work done on object = Fs Net force = ma So net work done on object = mas Equation of motion: $v^2 = u^2 + 2as$ For $u = 0$, $s = v^2/2a$ $= m((v^2 - u^2)/2)$ $= \frac{1}{2} mv^2$ (since $u = 0$) Work done = gain in kinetic energy So KE = $\frac{1}{2} mv^2$	
	(b)	(i)	Work done = power x time $= (37000)(11)$ $= 407 \text{ kJ}$ (3 sf) [need to show calculated raw value of 407] $= 410 \text{ kJ}$ (2 sf)(shown)	
		(ii)	Using $P = Fv$, and driving force = resistive forces at maximum speed. $(37000/2) = 1200v$ $v = 15.4 \text{ m s}^{-1}$ [Cannot assume the car travels for 11 seconds]	

2	(a)	(i)	Internal energy of a system is the sum of the random distribution of kinetic	
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